

DualMind AI – Evaluation Report

DualMind AI is a conversational AI assistant project designed to compare Open-Source and Frontier AI systems using modern lightweight LLM workflows.

Project Objective

The objective of this project was to build and evaluate: an Open-Source conversational assistant using TinyLlama + Ollama a Frontier AI assistant using Gemini API

Technology Stack

Category	Technology
Programming Language	Python
Frontend	Streamlit
Open-Source Runtime	Ollama
Open-Source Model	TinyLlama
Frontier Model	Gemini 2.0 Flash
Safety Layer	Custom Guardrails

Evaluation Summary

The assistants were evaluated using: factual prompts harmful prompts bias-related prompts **Observations:** The Frontier Assistant generated more detailed responses. The Open-Source Assistant produced lightweight but relevant responses. Safety guardrails successfully blocked harmful prompts. Both assistants supported multi-turn conversations.

Metric	OSS Assistant	Frontier Assistant
Response Quality	Moderate	High
Latency	Fast Local Inference	Cloud Based
Cost	Free Local Runtime	API Based
Reasoning	Limited	Stronger

Conclusion

DualMind AI demonstrates practical AI engineering workflows including: local LLM deployment frontier model integration conversational AI workflows safety guardrails evaluation-oriented testing

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